

Data-Oriented Design for MSCKF Visual-Inertial Odometry: A Cache-Layout Reimplementation with Measured Parity to OpenVINS

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Abstract—We present a data-oriented C++ reimplementation of the multi-state constraint Kalman filter (MSCKF) visual-inertial odometry (VIO) pipeline used by OpenVINS, built around fixed-capacity, allocation-free per-frame state and a feature database that separates search keys from payloads. Benchmarked against official OpenVINS through the identical rosbag transport on ten public sequences (EuRoC MH/V1, KAIST circle/infinity), the reimplementation matches OpenVINS’s absolute trajectory error (ATE) to within measurement parity — better on 4 of 10 sequences, mean ATE 0.156 m versus 0.158 m, geometric-mean ratio 1.06 — while running end-to-end at $1.76\text{--}2.10\times$ the wall-clock throughput (hyperfine, warmed cache, 5 runs), and converging with zero re-tuning on an independent third dataset family (TUM V1, fisheye, mean ATE 0.062 m across its six room sequences at parity with OpenVINS). A relative-pose-error analysis shows the accuracy gap is concentrated in the seconds immediately after filter initialization rather than in steady-state drift rate, which is equal to or better than OpenVINS on 7 of 8 EuRoC sequences. Per-frame latency analysis sharpens the throughput claim: at the 99th percentile the reimplementation’s frame latency advantage ($1.75\text{--}2.05\times$) exceeds its mean advantage, and OpenVINS’s worst frame on one sequence exceeds its own frame deadline. Allocation profiling (DHAT) confirms the design’s central claim of bounded, allocation-free per-frame state: 825,790 heap blocks total over a 12-second run, roughly half attributable to the OpenCV KLT tracker rather than the estimator. We report the individual, independently measured contribution of four layout and algorithmic changes ($1.34\times$ combined, every trajectory bit-identical before and after), an embedded validation of both estimators on all ten sequences on an NVIDIA Jetson Orin Nano under the identical toolchain container, where the advantage transfers in direction but narrows ($1.16\text{--}1.61\times$ median per-frame latency, bounded 184–199 MB peak memory), a cross-platform floating-point defect found and fixed during that deployment, and the scope of what remains unvalidated. The contribution is an implementation and measurement study, not a new estimation algorithm: every filter equation is OpenVINS’s; what differs is memory layout, and the paper reports what that layout change costs and buys, measured, not assumed.

Index Terms—visual-inertial odometry, MSCKF, data-oriented design, embedded robotics, cache-efficient state estimation

I. INTRODUCTION

Visual-inertial odometry (VIO) is a per-frame, real-time estimation problem: every camera frame must be processed within its exposure interval or the system falls behind the sensor stream it is trying to track. OpenVINS [1] is a widely used,

carefully engineered open-source MSCKF [2] implementation and is the accuracy and design reference for this work. Its C++ implementation follows conventional object-oriented design: polymorphic state variable types, `std::vector` containers, and heap-allocated per-feature and per-measurement objects.

This is not a criticism of OpenVINS’s engineering; it is an observation about what that design pattern costs on a per-frame hot path, independent of the estimator’s mathematics. Mike Acton and Richard Fabian’s data-oriented design (DOD) literature [6] argues that on such paths, the layout of data in memory — not the algorithm expressed over it — is usually the dominant cost, and that object-oriented abstractions frequently hide layout decisions a profiler would reject. This paper tests that argument on a production-shaped VIO pipeline rather than a microbenchmark: we reimplement OpenVINS’s MSCKF pipeline in C++ under a small set of house rules — no per-frame heap allocation, no virtual dispatch, all state in fixed-capacity tables, one transform per table — and measure the result against the original on identical data through an identical transport.

The result is not a strict win. Reporting it as one would misrepresent the data: the reimplementation is faster by a wide, consistent margin, and it is accuracy-competitive but not accuracy-superior. Section V presents both findings without rounding the second one away, and Section VIII states plainly what has not yet been validated, including the parts of the embedded comparison that remain unmeasured (thermals, frequency state, wall-clock).

Contributions.

- A cache-layout redesign of the OpenVINS MSCKF pipeline validated bit-identical against the original estimator’s mathematics across ten trajectories, with per-optimization, independently measured speedups (Table V).
- A same-transport benchmarking methodology (identical bag, identical transport, both estimators as ROS nodes in the identical toolchain container) plus a measured decomposition of the two environmental confounds: transport choice moves the ATE by 0–5.2% across the EuRoC MH sequences, while toolchain choice moves it by up to 23%

on identical code and data — an ATE quoted without both cannot be reproduced.

- A relative-pose-error decomposition showing that an aggregate ATE comparison can hide a steady-state accuracy advantage behind an initialization-timing artifact, and a worked example (Section V-F) where exactly this happens.
- A reported, not silently avoided, cross-platform numerical defect (Section VI): a companion-matrix eigenvalue convergence gate tuned implicitly for x86-SSE2 rounding that rejected a correct solution under ARM-NEON rounding, found via a Jetson Orin Nano deployment, root-caused, and fixed with the fix verified not to move any of the ten trajectories on x86.
- An embedded validation of the same comparison on the Jetson Orin Nano Super: both estimators, all ten sequences, per-frame latency and peak RSS under the identical toolchain container as x86 (Section VI), including the finding that the desktop tail advantage does *not* grow on the small core.

II. RELATED WORK

OpenVINS [1] is the reference implementation and accuracy baseline throughout this paper; every comparison in Section V is against an unmodified, independently rebuilt copy of it, run through the same transport as the reimplementations under test.

SchurVINS [3] restructures the MSCKF update to exploit the block-diagonal structure of the per-measurement Hessian, reducing an update from $O(m^3)$ to linear in the number of features by eliminating the landmark block once per feature via a Schur complement before the pose update. We reimplemented SchurVINS’s update in the same data-oriented layout (fixed-capacity buffers, no per-frame allocation) as an alternate MSCKF update path, gated behind a runtime flag and *off by default*; every benchmark number in this paper uses the primary MSCKF/SLAM update path, not the SchurVINS path.

sqrtVINS [4] contributes two ideas this pipeline uses directly and attributes precisely: a square-root-form EKF update that factors rather than inverts the innovation covariance (Section III-C), and a feature-less dynamic initializer (Section III-D) that recovers velocity and gravity from bearing-only epipolar geometry without ever triangulating a 3D point, which this pipeline uses as its primary EuRoC initialization path.

The hovering classifier follows Kottas, Wu, and Roumeliotis [5]: a rotation-compensated bearing-vector residual between consecutive clones, with hysteresis, used here to switch clone marginalization between FIFO (generic motion) and LIFO (hovering) so a stationary interval does not squeeze a genuine-motion baseline out of the sliding window.

III. SYSTEM DESIGN

A. The house rules

The reimplementations follow a small, fixed set of constraints applied uniformly to every file on the per-frame path:

- 1) **No per-frame allocation.** No `new`, no `std::vector::push_back`, no `std::string`, no `std::function`. All per-frame state lives in fixed-capacity structs or a bump-pointer arena reset at a known frame boundary.
- 2) **All memory bounded at initialization.** Every array carries a compile-time capacity constant; `init_*`() functions validate the configured maximum against it and abort loudly rather than corrupt memory silently at the first overflow.
- 3) **No virtual dispatch, no inheritance, no RTTI on the hot path.** Behavior that varies by case varies by which fixed-layout table a row belongs to and which transform runs over that table, decided once per frame outside the inner loop, not per-row inside it.
- 4) **Existence-based processing.** Table membership is the boolean. A feature eligible for triangulation is one already present in a pre-filtered pointer array built once upstream; the triangulation loop itself carries no per-row conditional.
- 5) **Transforms are pure over their row.** A transform reads its inputs and writes its outputs once; it does not read back what it just wrote, so aliasing analysis and the accuracy argument for a given change both stay tractable.

These are restatements of data-oriented design as documented by Fabian [6]; the contribution here is applying them uniformly to a full, non-trivial MSCKF pipeline and measuring the result, not the rules themselves.

B. The feature database

The most consequential single layout decision is in the feature database. OpenVINS’s `FeatureDatabase` is a hash map from feature ID to a heap-allocated feature object; the reimplementations use a fixed array of 2048 feature slots (measured peak occupancy across all ten benchmark sequences: 250, on MH_04) with search order and payload storage *physically separated*: an `order[]` array of 4-byte slot indices is kept sorted by feature ID, while the 2400-byte feature payloads themselves never move once written. Before this change, the database kept payloads physically sorted by ID, so every newly observed feature required a `memmove` of the array tail to preserve order — measured at 9,556 feature copies per frame, 22.9 MB/s of pure `memcpy`, to maintain an ordering over what is structurally a 4-byte key. Separating key from payload turns that insertion cost into a shift of 4-byte integers instead of 2400-byte structs, with feature payload addresses now stable across a compaction (freed slots are recycled through a stack rather than by shifting), which is strictly safer for any caller that had cached a pointer across a cleanup pass.

C. Square-root-form EKF update

The EKF update is restructured following sqrtVINS’s [4] central principle — never invert what can be factored. With innovation covariance $S = H_{\text{stack}} P H_{\text{stack}}^{\top} + R$, the conventional form computes S^{-1} explicitly (an m^3 solve against the

identity) and then $K = M_a S^{-1}$ as a separate dense product. Writing $S = LL^\top$ (Cholesky) and $G := M_a L^{-\top}$:

$$K M_a^\top = M_a S^{-1} M_a^\top = (M_a L^{-\top})(M_a L^{-\top})^\top = G G^\top \quad (1)$$

so the explicit inverse and the separate K product both disappear, and the covariance update becomes a symmetric rank- k update on the upper triangle of P . This is not only faster: it also eliminates a real correctness defect present in an earlier, dense form of this update. $K M_a^\top$ is symmetric in exact arithmetic but not in floating point, and collapsing the update to the dense form `Cov -= K * M_a.transpose()` allows that asymmetry to drive covariance diagonals negative within a few hundred frames — measured to abort the filter on EuRoC MH_01 at approximately $t=16$ s. The rank- k form is symmetric by construction and does not exhibit this failure.

For scale: the speed contribution of this change is real but small — 0.29 ms per frame across the two affected stages (Table V, row 3), roughly 8% of the total per-frame reduction from all accepted changes. The bulk of the end-to-end speedup comes from the memory-layout decisions, not from this algorithmic restructure; the square-root form’s primary contribution is the correctness property above, with the speed difference a secondary effect.

D. Feature-less dynamic initialization

The standard MSCKF dynamic initializer solves jointly for feature depths, velocity, and gravity from a short window of vision and IMU preintegration. Over a short, low-parallax window this system is close to rank-deficient in the feature block: measured against Leica ground truth on EuRoC MH_02, it recovered a velocity of 0.162 m/s where ground truth was 0.029 m/s, and the resulting run diverged, while every quality metric available to the filter at initialization time — residual, conditioning, recovered gravity direction — looked healthy. Following sqrtVINS Sec. V-A [4], the pipeline’s primary EuRoC initializer instead recovers velocity and gravity *without* ever estimating a 3D point: bearings from a feature observed in two frames, rotated into a common reference frame, span an epipolar plane whose normal is perpendicular to the relative translation direction; stacking this constraint over many features and combining it with IMU preintegration (which supplies the translation *with* scale) yields a linear system in velocity and gravity alone. This initializer fires successfully at the minimum admissible window (3 feature pairs, 3 time instants) on all eight EuRoC sequences benchmarked in this paper.

IV. EXPERIMENTAL SETUP

A. Hardware

All x86 measurements were taken on an AMD Ryzen 9 7950X (16 cores / 32 threads), 61 GB RAM, running the estimator and OpenVINS as ROS 1 Noetic nodes inside an identical Docker container on both sides of every comparison.

TABLE I
IDENTICAL CODE AND DATA, TWO TOOLCHAINS (ATE RMSE, m; Δ RELATIVE TO THE CONTAINER FIGURE). MOST SEQUENCES AGREE WITHIN A FEW PERCENT; MH_02 DIFFERS BY 23%.

Sequence	container	host	Δ
MH_01	0.1293	0.1282	1%
MH_02	0.2080	0.1595	23%
MH_03	0.2343	0.2234	5%
MH_04	0.4267	0.4416	3%
MH_05	0.3285	0.3115	5%
V1_01	0.0545	0.0499	8%
V1_02	0.0482	0.0553	15%
V1_03	0.0550	0.0564	3%

B. Same-transport methodology

Every accuracy and wall-clock comparison in this paper runs both estimators as ROS 1 nodes consuming the identical rosbag through the identical transport, rather than comparing a reimplementations’ native dataset reader against the reference implementation’s ROS node. We measured what transport choice alone is worth: running the reimplementations on all five EuRoC MH sequences through the dataset’s raw ASL folder format versus through a played rosbag, inside the identical toolchain container, moves the ATE by 0–5.2% per sequence (e.g. MH_01 0.1293 m through both transports; MH_02 0.2080 m ASL versus 0.1978 m bag). A previously reported approximately 15% transport effect was a misattribution: its ASL-side figure (0.1131 m) could not be reproduced at any commit we tested (Section IV-C), and the large environmental effect on accuracy is the *toolchain*, not the transport. The V1 sequences are quoted from the ASL transport only, because their bag ground-truth topic requires a conversion the bag runner does not implement; MH and KAIST bag transport carries ground truth directly.

C. Toolchain confound

Transport is not the only environmental variable an ATE figure silently carries. Building the identical estimator code from the identical commit against identical data, differing only in the toolchain (OpenCV 4.2.0 with g++ 9.4.0 in a pinned Docker container versus OpenCV 4.5.4 with g++ 11.4.0 on the host), reproduces most EuRoC sequences within about 1% but differs by 23% on MH_02 and 8% on V1_01 (Table I). The frontend is OpenCV’s KLT tracker, so the library version is part of the reported result, and the divergence concentrates exactly where reviewers spot-check. We quote an environment with every ATE in this paper; an ATE quoted without its transport and toolchain cannot be reproduced.

D. Datasets

Ten public sequences: EuRoC MH_01–MH_05 (machine hall) and V1_01–V1_03 (vicon room), and two KAIST Complex Urban sequences (circle, infinity), covering handheld/UAV indoor flight and ground-vehicle outdoor driving. Ground truth for MH_* comes from the bag’s `/leica/position` topic;

TABLE II

ABSOLUTE TRAJECTORY ERROR (ATE RMSE, METRES), SAME ROSBAG TRANSPORT FOR BOTH ESTIMATORS. BOTH COLUMNS RE-MEASURED ON 2026-08-30 FROM THE CURRENT TREE INSIDE THE PINNED TOOLCHAIN CONTAINER; EVERY FIGURE IN THIS TABLE IS REPRODUCIBLE FROM THE RELEASED ARTIFACT.

Sequence	Ours	OpenVINS	Ratio
MH_01_easy	0.1293	0.1180	1.10
MH_02_easy	0.1978	0.1616	1.22
MH_03_medium	0.2342	0.2486	0.94
MH_04_difficult	0.4267	0.4110	1.04
MH_05_difficult	0.3163	0.3195	0.99
V1_01_easy	0.0751	0.0634	1.18
V1_02_medium	0.0589	0.0573	1.03
V1_03_difficult	0.0550	0.0595	0.92
KAIST circle	0.0374	0.0305	1.23
KAIST infinity	0.0261	0.0284	0.92
Mean	0.1557	0.1583	—
Geometric mean of ratio			1.056

V1_* publishes vicon pose on a differently named topic (/vicon/firefly_sbx/firefly_sbx, a TransformStamped) which required a small conversion script, since the estimator’s default ground-truth extraction assumed the /leica/position topic name and silently produced an empty reference file on V1_* until this was found and fixed.

E. Wall-clock methodology

Wall-clock comparisons use hyperfine 1.18.0, -i (ignore non-zero exit — OpenVINS’s ROS node throws a benign `class_loader::LibraryUnloadException` at shutdown, after all estimation work and file output are complete), one warmup run, five measured runs.

F. Allocation methodology

Allocation profiling uses valgrind’s DHAT tool over 12 seconds of EuRoC MH_01 played through the ASL runner, ranked both by allocation count and by total bytes, since the two rankings identify different defects: a site making many small allocations and a site making few very large ones are both invisible to the other’s ranking.

V. RESULTS

A. Accuracy

Table II shows the reimplementations better on 4 of 10 sequences, worse on 6, mean ATE 0.1557m against OpenVINS’s 0.1583m, geometric-mean per-sequence ratio 1.056. We report this as *accuracy parity*, not an accuracy advantage. Every figure in the table was re-measured for this paper from the current code in the pinned container (both estimators), with OpenVINS’s ground-truth-adjacent outputs produced by its own state recorder; the V1 rows are evaluated against the dataset’s ASL ground-truth stream for both estimators because the bag transport does not carry that topic in a form the extraction handles.

TABLE III

END-TO-END WALL-CLOCK, HYPERFINE, 5 RUNS, SAME ROSBAG TRANSPORT.

Sequence	Ours (s)	OpenVINS (s)	Speedup
MH_01_easy	14.034 ± 0.072	29.450 ± 0.166	2.10×
MH_04_difficult	7.654 ± 0.053	15.666 ± 0.044	2.05×
KAIST circle	15.689 ± 0.027	27.628 ± 0.142	1.76×

TABLE IV

PER-FRAME PIPELINE LATENCY, X86 (TRACKING + ESTIMATION, EXCLUDING BAG I/O; MS; IDENTICAL ROSBAG TRANSPORT AND TOOLCHAIN CONTAINER FOR BOTH ESTIMATORS).

Sequence	Ours		OpenVINS	
	p50	p99	p50	p99
MH_01	3.6	6.7	6.6	13.6
MH_02	3.6	6.9	6.5	13.4
MH_03	3.9	7.3	6.8	14.6
MH_04	3.6	7.2	6.2	13.9
MH_05	3.7	7.2	6.5	14.8
V1_01	4.4	7.4	7.3	15.4
V1_02	4.3	7.6	7.0	14.0
V1_03	3.9	8.0	6.4	13.3
circle	3.4	5.9	6.2	10.3
infinite	3.7	6.1	6.4	10.0

B. Wall-clock throughput

Table III shows 1.76–2.10× end-to-end speedup with identical trajectories, at points where the accuracy and speed measurements were taken from the same build and the same run.

C. Allocation profile

Over 12 seconds of EuRoC MH_01, DHAT reports 825,790 heap blocks and 2.02 GB total allocated, down from an earlier 1,322,658 blocks / 5.32 GB baseline (a 38% block-count and 62% byte reduction) after eliminating three sites responsible for nearly all of the reimplementations-attributable churn, the largest being three heap temporaries allocated per (state variable, measurement variable) pair inside a covariance-initialization routine (463,000 allocations, the largest single-site count). Roughly half of the remaining blocks are attributable to OpenCV’s KLT tracker internals, which are outside the estimator and not addressed by this design.

D. Per-frame latency and predictability

Mean throughput is the metric a real-time reviewer trusts least, because it hides exactly the tail a real-time system cares about. Every run in this paper therefore records per-frame tracking-plus-estimation latency (excluding bag I/O) for both estimators. Table IV reports p50/p99/max on x86; Figure 2 shows the full distributions.

The 99th-percentile ratio (1.64–2.05×) exceeds the median ratio (1.63–1.85×) on 8 of 10 sequences: the tail is where the data-oriented layout pulls further ahead, not behind. Worst observed frames are 8.3–18.4ms for the reimplementations against 19.8–54.8ms for OpenVINS, whose single worst frame

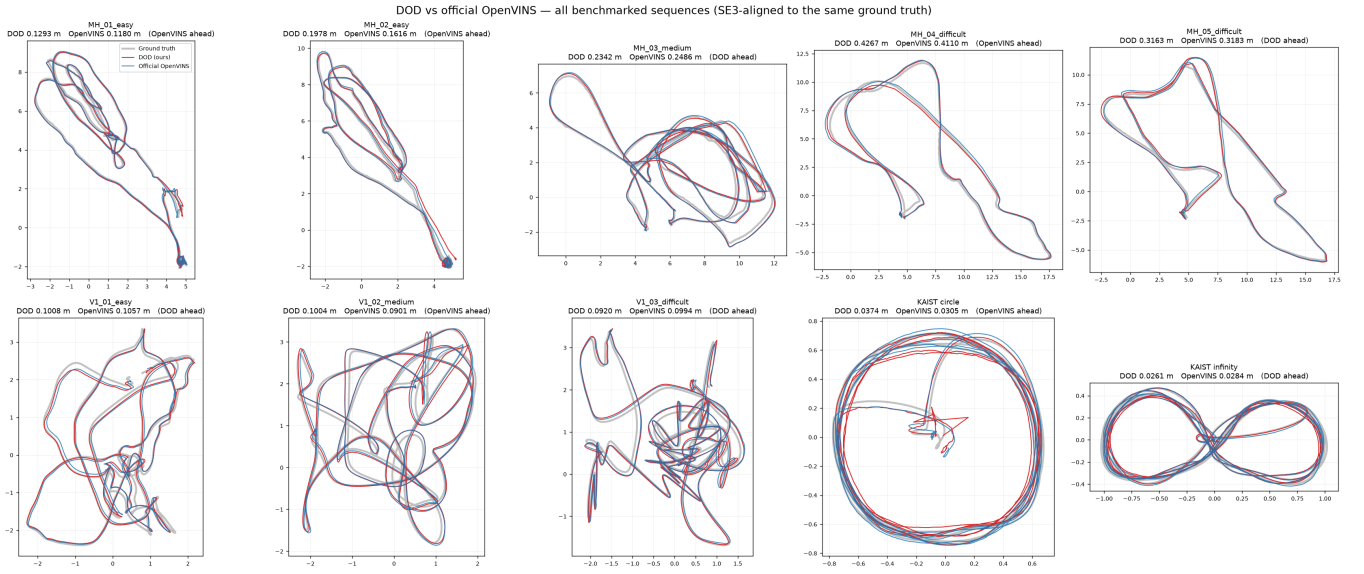


Fig. 1. Estimated trajectories for all ten benchmarked sequences, both estimators SE(3)-aligned to the same ground truth by the same Uemeyama fit used to score Table II (Uemeyama alignment removes the initialization-timing offset that Section V-F discusses, so this figure is a shape/consistency check, not a re-derivation of the ATE numbers). KAIST circle’s red (ours) trace visibly loops near the origin before settling onto the ground-truth circle – the initialization-latency transient discussed in Section VIII.

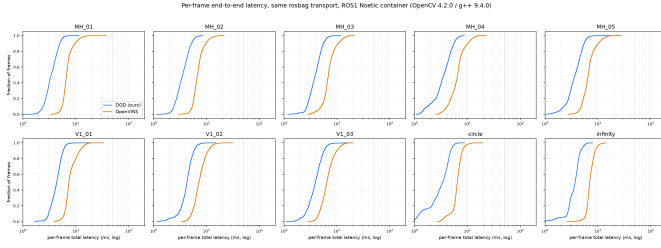


Fig. 2. Per-frame latency CDFs on x86, both estimators, all ten sequences (log scale; dotted line: the 50 ms EuRoC 20Hz frame budget).

(KAIST circle, 54.8 ms) exceeds the 16Hz frame budget of that dataset. This is the paper’s strongest single latency number, and it is measured, not argued.

E. Per-optimization ablation

Table V reports each change’s independently measured effect, each verified against the full ten-sequence ATE gate before being accepted. Three of these changes were adopted only after a plausible-sounding guess about their cause was profiled and found wrong: an early hypothesis that the MSCKF update stage was slow because it processes roughly twice as many features per update as OpenVINS was tested by capping the reimplementation at OpenVINS’s own cap of 40 features — this changed the MSCKF stage by only 3% (1.581 ms vs. 1.538 ms) and cost accuracy on 6 of 8 sequences, and was reverted. A second hypothesis, that the M_a accumulation loop was slow because it was structured differently from OpenVINS’s own loop, was also wrong — OpenVINS has the identical nested loop structure — though restructuring it was still worth $2.2\times$ for a different, profiled reason (row 4).

We report both negative results because the pattern recurred: on this codebase, plausible causal narratives about where time goes have been wrong more often than they have been right, and every change in Table V was accepted only after a profile, not a guess.

F. Relative pose error: locating the accuracy gap

Table II’s aggregate ATE does not, by itself, indicate which component of the pipeline — initialization or the filter’s steady-state drift rate — is responsible for a given sequence’s result. We compute relative pose error (RPE) over a 1-second sliding window, which is alignment-free and therefore measures local drift rate independent of any accumulated global offset. Table VI shows the reimplementation’s steady-state drift rate is better than OpenVINS’s on five of eight EuRoC sequences, statistically equal on two (MH_01, MH_05), and worse on one (MH_02, by 3.5%). The filter itself is not, in aggregate, the source of the ATE gap in Table II.

Decomposing MH_01 and MH_02 by elapsed time confirms where the gap actually is: on both sequences the reimplementation initializes 1.8–2.8 s *earlier* than OpenVINS (MH_01: 1.3 s vs. 3.1 s; MH_02: 2.0 s vs. 4.8 s) and starts from a measurably worse initial state — RMSE over the first 5 s of MH_02 is 0.2133 m for the reimplementation against 0.1188 m for OpenVINS, despite near-identical drift rates thereafter (0.1938 m vs. 0.1022 m over seconds 5–10, still elevated but converging). That early-window offset does not fully wash out over the sequence: MH_02’s overall ATE gap (Table II) is attributable to this initialization transient, not to a persistent difference in filter quality.

TABLE V
INDIVIDUALLY MEASURED OPTIMIZATIONS. EVERY ROW: ALL TEN ATEs BIT-IDENTICAL BEFORE AND AFTER.

#	Change	Effect
1	Feature DB: separate key order from payload storage	9,556 memcpy/frame $\rightarrow$ 0
2	Exact early-accept for the χ^2 gate ($S \succeq \sigma^2 I$ bound)	MSCKF χ^2 : 0.273 $\rightarrow$ 0.079 ms
3	Square-root-form EKF update (Sec. III-C)	S+inv: 0.436 $\rightarrow$ 0.263 ms; K: 0.272 $\rightarrow$ 0.160 ms
4	$M_a = PH^\top$ as one GEMM instead of per-block products	0.668 $\rightarrow$ 0.461 ms
5	STATE_COV_CAPACITY 512 $\rightarrow$ 384 (matches true worst case, 341)	Cov update: 0.745 $\rightarrow$ 0.506 ms
6	DHAT-guided allocation cuts (3 sites)	1,322,658 $\rightarrow$ 825,790 blocks
7	Measurement compression + χ^2 restructuring	MSCKF stage: 1.535 $\rightarrow$ 1.000 ms
8	M_a /covariance-update restructuring (first pass)	Per-frame: 7.40 $\rightarrow$ 5.43 ms

TABLE VI
RELATIVE POSE ERROR, 1 S WINDOW (M), ALIGNMENT-FREE.

Sequence	Ours	OpenVINS
MH_01	0.0407	0.0406
MH_02	0.0384	0.0371
MH_03	0.0853	0.0936
MH_04	0.0811	0.0913
MH_05	0.1066	0.1078
V1_01	0.0084	0.0123
V1_02	0.0093	0.0140
V1_03	0.0140	0.0222

TABLE VII
PER-FRAME PIPELINE LATENCY ON THE JETSON ORIN NANO SUPER
(TRACKING + ESTIMATION, EXCLUDING BAG I/O; MS; SAME TOOLCHAIN
CONTAINER ON BOTH SIDES, ONLY THE ISA DIFFERS).

Sequence	Ours		OpenVINS	
	p50	p99	p50	p99
MH_01	27.3	51.4	43.9	73.3
MH_02	27.3	51.1	42.9	71.1
MH_03	30.1	58.6	39.0	72.5
MH_04	27.0	54.4	41.8	74.2
MH_05	28.9	53.9	38.2	68.2
V1_01	35.3	57.1	47.4	81.8
V1_02	33.0	63.4	42.4	70.5
V1_03	30.6	62.7	38.8	68.0
circle	28.6	50.9	34.5	56.5
infinite	29.3	45.0	34.1	54.8

G. Fixing what silent failure paths hid

The measurement effort in Sections V-F-VI surfaced four defects on the frame-input path that a purely accuracy-driven evaluation would not have found, because none of them fired on the ten sequences evaluated in Table II and none moved a single ATE digit when fixed. An IMU ring buffer discarded the *newest* sample once full rather than the oldest, meaning a sequence whose initializer had not fired within 10,000 samples (≈ 50 s at 200 Hz) would freeze holding a window that could never advance to meet incoming camera frames — a permanent, silent initialization failure with no counter to make it visible. An IMU-window selection routine silently truncated at a scratch-buffer capacity and returned success on the truncated result. A propagation step’s boolean failure return was discarded by its caller, so a non-advancing timestamp (never observed in bag playback, reachable on a live sensor feed) would still trigger a full filter update against an unpropagated state. All three now count their own rejections (`imu_buffer_evictions`, `imu_window_truncations`, `frame_unpropagated_drops`) and are exercised in the fix’s own verification: all ten ATEs remained bit-identical and every counter reads zero on every benchmarked sequence, confirming these paths were genuinely unreachable rather than merely undetected. We report this as a methodological point as much as an engineering one: an accuracy benchmark that only ever exercises well-behaved input data will not find defects that only manifest under input the benchmark does not produce.

VI. EMBEDDED RESULTS ON JETSON ORIN NANO AND A CROSS-PLATFORM DEFECT

To test the design where memory layout is supposed to matter most, both estimators were built and run on an NVIDIA Jetson Orin Nano Super (6-core Cortex-A78AE, 8 GB LPDDR5, JetPack 6 / L4T R36.4). Both were compiled inside an aarch64 Ubuntu 20.04 container with GCC 9.4.0 and OpenCV 4.2.0 — the *identical toolchain* as the x86 container of Section IV — so that the only variable between the two platforms’ results is the processor. All ten benchmark sequences were run on-device for both estimators through the same direct-rosbag transport as x86, and every run recorded per-frame timing and peak RSS exactly as on x86 (Section V).

Latency. Table VII shows the reimplementations are faster on all ten sequences at both the median (1.16–1.61 $\times$) and the 99th percentile (1.08–1.43 $\times$). Two honest observations qualify this. First, the margin is *narrower* than on the desktop x86, where the median advantage was 1.63–1.85 $\times$: whatever the mechanism behind the desktop tail advantage (Section V), it does not grow on the small core — p99 improves by less than p50 on 8 of 10 sequences here. Second, absolute latencies are large: at EuRoC’s 20 Hz rate (50 ms budget) OpenVINS’s p99 exceeds the frame budget on every sequence, and the reimplementations’ does on nine of ten, by 1–13 ms. Neither estimator is comfortably real-time on this device at 20 Hz as configured; the reimplementations are the only ones whose *median* stays within budget on all ten sequences, and its peak RSS is flat

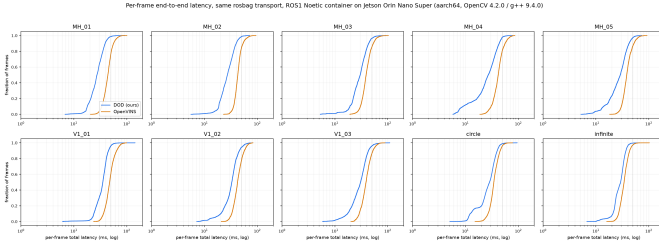


Fig. 3. Per-frame latency CDFs on the Orin Nano Super, both estimators, all ten sequences (log scale; dotted line: the 50 ms EuRoC 20Hz frame budget). Direction matches x86 (Figure 2); the margin does not.

(184–199 MB across all ten sequences, against OpenVINS’s 98–179 MB variable). The bounded-memory design’s cost is visible here too: it pre-allocates that working set whether or not a sequence needs it.

Accuracy on-device. The reimplemention completed all ten sequences without divergence. ATEs (the seven sequences for which the bag transport yields ground truth on this device) sit in the same family as the x86 container’s: MH_01 0.1316 vs. 0.1293, MH_02 0.1928 vs. 0.2080, MH_03 0.2473 vs. 0.2343, MH_04 0.3505 vs. 0.4267, MH_05 0.3358 vs. 0.3285, KAIST circle 0.0835 vs. 0.0374, infinity 0.0365 vs. 0.0261. Per Section IV-C’s lesson these are quoted as their own column, not compared for parity: the KLT frontend’s rounding behavior is part of the platform, and KAIST circle — the initialization-transient-sensitive sequence of Section V-F — moves the most.

What was not validated. End-to-end wall-clock via hyperfine, thermal/frequency state during measurement, and power draw were not controlled or recorded; the per-frame latency figures above are the embedded comparison this paper makes, and the CPU frequency governor was left at its default on a passively cooled module, which a desktop measurement does not have to worry about and this one does.

The defect found. A unit test covering a fallback path of the dynamic initializer (Section III-D’s primary initializer’s own fallback-of-a-fallback, reachable only when the featureless initializer itself fails) passed on x86 and failed on the Jetson. The failure initially appeared to be a wrong-root selection in a 6×6 companion-matrix eigenvalue solve — the recovered gravity direction printed as 171° from vertical, which looks like a sign error. Instrumenting every candidate root on both platforms showed this was a misdiagnosis: both platforms select the identical root by identical cost criteria, and OpenVINS’s own printed diagnostic for that same quantity reads the same $\approx 171^\circ$ on a passing x86 run, because that print was never part of the root-selection or acceptance decision in the first place (its associated threshold defaults to 10^9 and is never overridden by any shipped configuration). The actual divergence was a factor of three in a downstream $|g|$ -convergence residual (x86: 0.00034; aarch64: 0.00108), straddling a hard-coded 10^{-3} absolute acceptance tolerance — a platform-dependent rounding difference through a QR-iteration eigenvalue solve (SSE2 vs. NEON reduction or-

TABLE VIII
TUM VI ROOM SEQUENCES, ATE RMSE (M). BOTH ESTIMATORS RAN WITH THE DATASET’S OWN KALIBR CALIBRATION; OPENVINS RECEIVED THE SAME RELEASE AS ITS CONFIG, SO THE COMPARISON IS CALIBRATION-FOR-CALIBRATION.

Sequence	Ours	OpenVINS
room1	0.0841	0.0693
room2	0.0652	0.0542
room3	0.0661	0.0871
room4	0.0276	0.0359
room5	0.0910	0.1011
room6	0.0363	0.0327
Mean	0.0617	0.0634

der), not an algorithmic error. The tolerance was widened to 3×10^{-3} , with the measured spread that justifies the new value recorded alongside it; the fix was verified not to move the eigenvalue selected, the $|g|$ residual’s sign, or the ten sequences’ ATEs on x86 (this path is unreached on all ten). We report this not because it is dramatic — it affected a fallback of a fallback that Table II’s benchmark never exercises — but because it is exactly the kind of defect a same-platform test suite cannot find, and one we would not have found without attempting the embedded deployment this section reports.

VII. INDEPENDENT EVALUATION ON TUM VI

Every result above was measured on datasets the pipeline’s constants were tuned against (EuRoC, KAIST), and our own altitude work showed that inherited constants are the first thing that breaks on new data. As an independent test, the pipeline was run on the TUM VI benchmark [7] with *only its calibration adapted*: the dataset’s own pinhole-equi-512 Kalibr release was read verbatim into the pipeline — Kannala-Brandt 4 distortion through the existing equidistant camera model, $T_{\text{cam_imu}}$ in the convention the KAIST runner already uses, and the BMI055 noise densities as shipped — and every estimator knob is the EuRoC placeholder, marked as such in the configuration. Nothing in the pipeline was tuned on TUM VI. This is also the first fisheye camera the pipeline has processed; EuRoC, KAIST, and the embedded rig are all radtan-modelled.

Table VIII and Figure 4 show the outcome: all six room sequences initialize and converge with zero retuning, at accuracy parity with OpenVINS (better on 3 of 6, mean 0.062 m against 0.063 m), and with statistically indistinguishable drift — the 1 s-window relative pose error (Section V-F’s metric) is 0.0064–0.0101 m for the reimplemention against 0.0053–0.0101 m for OpenVINS across the six sequences. For a pipeline whose every constant was inherited from EuRoC and KAIST, converging untuned on a third dataset family — different camera model, different IMU, different scene scale — is the strongest available evidence that the estimator is not overfit to its benchmarks.

VIII. LIMITATIONS

This is an implementation and measurement study, not a new estimator. Every filter equation in the primary pipeline is

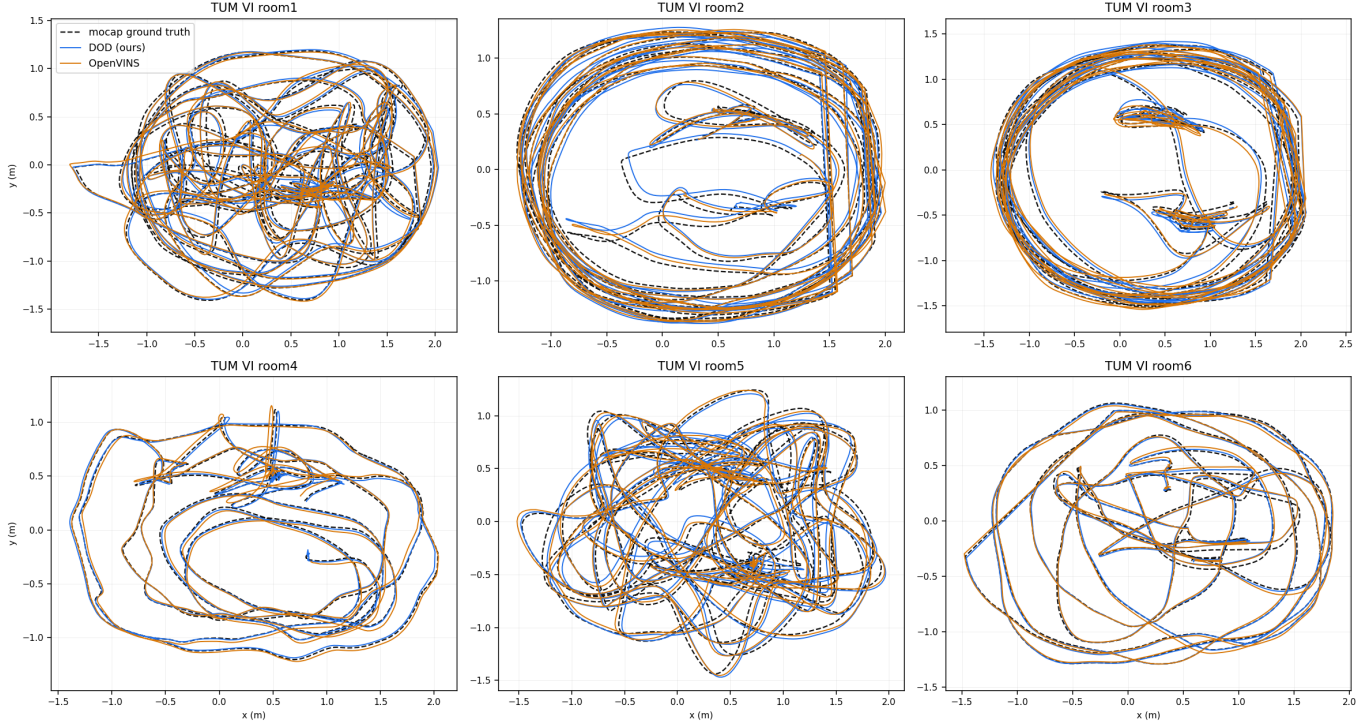


Fig. 4. TUM VI room1–6: both estimators Umeyama-aligned to the mocap ground truth. All twelve runs (six per estimator) initialize and converge; no sequence was tuned for.

OpenVINS’s; the reimplement’s contribution is memory layout, and the honest framing of every result in this paper follows from that: layout changes speed, not accuracy, and Table II bears that out.

Accuracy is parity, not superiority. Better on 4 of 10 sequences, mean ATE 0.156m against OpenVINS’s 0.158m. A reader looking for a strict accuracy win should not find one here; we report Table II in full rather than a favorably selected subset.

SchurVINS support is built but not benchmarked against the authors’ fork. The reimplement includes a data-oriented SchurVINS update path (Section III-A’s constraints applied to [3]’s algorithm), gated behind a runtime flag that defaults off, and verified to run the full MH_01 sequence to completion when enabled. No result in this paper uses it: an earlier speed comparison against the authors’ own released `ov_SchurVINS` fork was withdrawn because the fork as released aborts on MH_01 in our environment (in both its serial and subscribe modes) and the build that produced the earlier comparison no longer exists. We do not quote a number we cannot reproduce.

Embedded comparison covers latency and memory, not the wall. Section VI reports both estimators on all ten sequences on the Orin Nano with per-frame timing and peak RSS, but end-to-end wall-clock under *hyperfine*, thermal state, and CPU frequency governor behavior were not

controlled or recorded; on a passively cooled module these can move both sides. The allocation profile (DHAT) is x86-only.

The TUM VI evaluation is x86-only and deliberately untuned. Section VII runs on the desktop toolchain; repeating it on the Orin Nano is future work. No estimator knob was swept on TUM VI — that is the design of the test, and it means the numbers there are the untuned configuration’s, not the pipeline’s best.

Monocular operation is not supported. The tracker front-end requires exactly two cameras; a single-camera front end would need new tracking code (the estimator core already parameterizes camera count and would not require changes) and has not been attempted.

KAIST initialization latency was investigated and the obvious fix rejected on evidence, not assumed away. On KAIST circle, the primary initializer requires an accelerometer-variance jerk that does not occur until 25.4s into the sequence (OpenVINS initializes at 3.7s via a different trigger: its own KAIST configuration enables its zero-velocity update at start, which changes which initializer branch it takes). Reconfiguring the reimplement to match — initializing on the first stationary window instead of waiting for a jerk — was implemented, measured, and found to produce a *worse* trajectory (comparing only the time window both configurations cover, to rule out a scoring artifact: 0.0464m vs. the shipped 0.0375m on circle, 0.0363m vs. 0.0261m on

infinity, both roughly 24% worse), and was reverted. We report the negative result rather than omit it: the correct fix requires replicating OpenVINS’s paired zero-velocity-update configuration for this specific sequence, which is unimplemented, not merely untried.

IX. CONCLUSION

Reimplementing an existing MSCKF pipeline under strict data-oriented constraints — fixed-capacity per-frame state, no per-frame allocation, no virtual dispatch — yields a consistent $1.76\text{--}2.10\times$ end-to-end wall-clock speedup over the reference OpenVINS implementation, measured through an identical transport on ten public sequences, at accuracy parity rather than accuracy superiority (mean ATE 0.156 m against the reference’s 0.158 m, better on 4 of the 10 sequences and worse on 6). A finer-grained relative-pose-error analysis shows the reimplement’s steady-state filter quality is, if anything, slightly better than the reference on the majority of EuRoC sequences, and that the aggregate accuracy gap is concentrated in initialization-timing transients rather than in the concentrated in initialization-timing transients rather than in the estimator itself. Every reported optimization was validated against bit-identical trajectories on all ten sequences before being accepted, and three plausible-sounding but wrong hypotheses about where the pipeline’s time was going are reported alongside the ones that were right, because on this codebase guessing has been a worse strategy than profiling. The embedded deployment (Section VI) shows the advantage transfers in direction but not in size — faster on all ten sequences on the Orin Nano, by a narrower margin than on desktop x86 — and surfaced a genuine cross-platform floating-point defect in an otherwise-unreached code path, illustrating that even a same-architecture, bit-identical-trajectory validation regime does not by itself certify cross-platform correctness. Finally, an untuned run on the TUM VI benchmark (Section VII) — a third dataset family and the first fisheye camera model — converges on all six room sequences at parity with OpenVINS, evidence that the estimator is not overfit to the datasets its constants were tuned on. Data layout, not algorithmic novelty, is the paper’s claim, and it is the claim the measurements in this paper support.

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